

PRACTICAL-07

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int global_var = 100;
static int static_var = 200;
void show_addresses()
{
    int local_var = 300;
    printf("\n===== MEMORY ADDRESSES =====\n");
    printf("Code/Text address  : %p\n", (void *)show_addresses);
    printf("Global variable   : %p\n", (void *)&global_var);
    printf("Static variable    : %p\n", (void *)&static_var);
    int *heap_var = malloc(sizeof(int));
    if (heap_var == NULL)
    {
        printf("Heap allocation failed\n");
        return;
    }
    *heap_var = 400;
    printf("Heap variable      : %p\n", (void *)heap_var);
    printf("Stack variable     : %p\n", (void *)&local_var);
    printf("\nProcess ID (PID)   : %d\n", getpid());
    free(heap_var);
}
int main()
{
    show_addresses();

    printf("\nPress Enter to exit...\n");
    getchar();
    return 0;
}
```

Screen Shots:

```
sri_charan@SRICHARAN: ~/Pi  × + v
sri_charan@SRICHARAN:~$ ls
OSLab OSLab2 Practical 'Practical 1' Skill hello.c
sri_charan@SRICHARAN:~$ cd Practical
sri_charan@SRICHARAN:~/Practical$ gcc Practical7.c -o Practical7
sri_charan@SRICHARAN:~/Practical$ ./Practical7

===== MEMORY ADDRESSES =====
Code/Text address : 0x5ee9e8b6e209
Global variable   : 0x5ee9e8b71010
Static variable   : 0x5ee9e8b71014
Heap variable     : 0x5eea0f9d6020
Stack variable    : 0x7ffe71d06ffc

Process ID (PID)  : 445

Press Enter to exit...
sri_charan@SRICHARAN:~/Practical$
```